

RESEARCH ARTICLE

Integration of Extended Kalman Filtering and Deep Reinforcement Learning for Autonomous UAV Navigation Under GPS Jamming

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ABSTRACT This paper presents a framework for autonomous unmanned aerial vehicle (UAV) navigation in environments subject to Global Positioning System (GPS) jamming. The framework conceptualizes interference as both a security threat and a source of actionable information. The proposed method utilizes the predictable effects of jamming on received power and path loss to infer environmental parameters critical for navigation. An extended Kalman Filter (EKF) estimates transmission power, received signal strength, and path loss exponent. These estimates are integrated into a Deep Reinforcement Learning (DRL) algorithm, which enables the UAV to optimize its trajectory by combining signal observations with adaptive navigation policies. This approach reduces the adverse effects of jamming and extracts useful parameters for navigation. Simulation results indicate that the EKF-DRL framework supports robust and accurate navigation under severe interference, enabling UAVs to complete missions reliably.

INDEX TERMS UAV, GPS, jamming, GPS denied, path planning, DRL, EKF.

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs), also known as drones, have become a key technology in disaster response, environmental monitoring, logistics, and intelligent sensing [1], [2], [3]. Their flexibility and cost-effectiveness make them valuable for emergency rescue operations, rapid information delivery, and situational awareness. UAVs can also be deployed quickly as aerial communication platforms. They can operate as airborne base stations and support various mission-critical operations.

Recent advances have transitioned UAVs from manual operation to fully autonomous systems, equipped with capabilities such as obstacle avoidance, trajectory tracking, and adaptive mission planning [4], [5]. While these enhancements improve efficiency and expand applications, they also introduce new security risks. UAVs rely on open wireless channels and broadcast signals, making them particularly vulnerable to malicious interference. Among these threats, GPS jamming is especially severe [6]. In such attacks, adversarial transmitters

emit interference that disrupts satellite signals, causing position and timing errors. These disruptions can hinder mission execution or prevent UAVs from navigating to their intended destinations. GPS jamming is thus a highly effective method for denying UAV access to sensitive areas, critical infrastructure, or restricted airspace [7].

Traditional strategies for UAV navigation often depend on frequent map reconstruction and trajectory re-planning. However, these methods incur significant computational overhead and perform poorly in complex or dynamic environments, especially under adversarial conditions. To address these limitations, researchers are increasingly turning to AI-based approaches, particularly DRL. DRL has demonstrated strong capabilities in solving problems that are intractable for classical optimization techniques, driving the development of intelligent path-planning frameworks. As a result, DRL is now widely regarded as a promising solution for enabling robust UAV navigation in the presence of GPS jamming attacks.

In this paper, we propose a novel framework for UAV navigation under GPS jamming conditions. Unlike traditional approaches that treat interference solely as a security threat,

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our method leverages it as a source of actionable information. Specifically, we exploit the predictable impact of jamming on received power and path-loss characteristics to infer environmental parameters that support navigation. An EKF is employed to estimate critical jamming-related parameters, including transmit power, received signal strength, and the path-loss exponent. These estimates are then integrated into a DRL-based decision-making process, enabling the UAV to adapt its trajectory by combining physical-layer signal observations with learned navigation policies. This dual-purpose strategy both mitigates the adverse effects of GPS jamming and exploits interference for parameter estimation, forming the core contribution of our work. Simulation results demonstrate that the proposed EKF-DRL framework achieves resilient and intelligent UAV navigation, allowing UAVs to maintain reliable mission performance and accurately reach target destinations, even in the presence of strong jamming signals.

The remainder of this paper is structured as follows. Section II reviews the related work. Section III introduces the system model and underlying assumptions. Section IV presents the EKF-based estimation framework for transmitter power and the path-loss exponent. Section V discusses conventional reinforcement learning methods and details the proposed DRL-based navigation framework for UAVs in GPS-jamming environments, with particular emphasis on the integration of EKF and DRL as well as the convergence analysis. Section VI reports the experimental results and provides an in-depth discussion. Finally, Section VII concludes the paper.

II. RELATED WORK

Recent research has increasingly combined artificial intelligence (AI) methods with estimation techniques to improve UAV navigation in GPS jamming environments. In particular, several studies have integrated DRL with the EKF to enhance path planning under GPS jamming conditions [8], [9], [10]. In [11], a reinforcement learning-aided EKF framework is presented that eliminates the need for manual filter tuning, thereby ensuring robustness against inaccurate initializations and filter gain uncertainties. Similarly, [12] introduced two nonlinear EKF-based filtering schemes tailored for fixed-wing UAVs, which effectively capture nonlinear flight dynamics and improve attitude estimation performance.

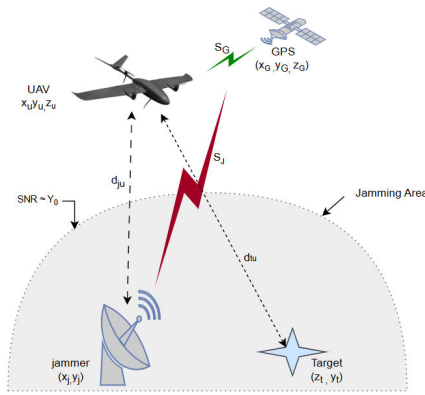
In addition to these estimation-focused approaches, efforts have also been directed toward exploiting jammer signals for navigation and mission continuity. In [13], an autonomous jamming-assisted navigation system was developed, where proportional-integral-derivative (PID) control was employed to estimate jammer power and position, subsequently enabling optimized trajectory planning toward mission objectives. In parallel, [14] the authors proposed the Reinforcement Learning-based Decision-Making for Unmanned Aerial Vehicle (RLDM-UAV) model to address GPS spoofing, wherein Deep Q-Network (DQN) were enhanced with an experience replay mechanism prioritizing critical transitions.

This approach allowed UAVs to jointly detect and mitigate spoofing attempts while adapting to sophisticated adversarial strategies.

Expanding from individual UAV solutions to collective behavior, at the swarm level, anti-jamming strategies have been studied using multi-agent frameworks. In [15], the multi-UAV anti-jamming problem was formulated as a decentralized partially observable Markov decision process (POMDP) and solved via multi-agent reinforcement learning (MARL), enabling distributed and cooperative defense strategies. Likewise, [16] improved UAV swarm communication under jamming by integrating reinforcement learning with multi-parameter programming, where decision-making considered a multi-dimensional parameter space including frequency, motion, and antenna configurations. Beyond anti-jamming and navigation, complementary approaches have focused on path planning and resource allocation. In [17], a radio-map-assisted planning algorithm was presented, relying on signal power and Signal-to-Interference-plus-Noise Ratio (SINR) maps that incorporated terrain-induced losses and interference effects to support trajectory optimization. Similarly, [18] studied UAV deployment within a congestion game framework, where joint optimization of transmission power and deployment positions was achieved through analysis of UAV utility functions and Nash equilibrium conditions. For large-scale UAV networks, [19] proposed a mean-field game approach to power control optimization, ensuring scalability and computational efficiency in dense deployments. In another direction, [20] modeled UAV jammers and ground users as intelligent agents, where the jammer optimized its three-dimensional trajectory using deep recurrent Q-networks (DRQN), while ground users adopted adaptive countermeasures based on Markov decision process formulations. Other work has investigated the use of UAVs to mitigate the effects of jamming [21], where UAVs are deployed as aerial hot zones to enhance SINR and restore downlink performance, while also highlighting that UAVs may cause significant uplink interference due to line-of-sight propagation, requiring mitigation through coordination and power control. Relatedly, [22] proposed DET-FAIR-WINGS, which combines multi-agent reinforcement learning and transformer-based detection to improve UAV-Base Station link reliability under attacks. The approach adaptively manages discrete parameters, such as retransmissions and bandwidth allocation, and continuous parameters, like beam angles and elevations, achieving faster convergence, lower latency, and reduced packet loss in 5G networks. Reference [23] presented an RSS-assisted Artificial Potential Field (RCA-APF) algorithm for UAV path planning under cyber and GPS attacks. It combines trilateration-based localization with potential field control to maintain navigation when GPS sensors fail. While these studies focus on trajectory planning, power control, and attack mitigation under standard navigation, our work addresses UAV navigation in GPS-denied environments by integrating EKF-based signal estimation with DQN

TABLE 1. Comparison of GPS jamming and spoofing countermeasures for UAV navigation. Abbreviations: POMDP – Partially Observable Markov Decision Process; PID – Proportional–Integral–Derivative.

Refs.	Problem	Techniques Considered						Multi-UAV	Methodology
		EKF	RL/DRL	Game Theory	Path Plan	Power Ctrl	Detect/Mit.		
[11]	GPS jamming	✓	✓	–	–	–	–	–	RL–EKF estimator.
[12]	Attitude estimation	✓	–	–	–	–	–	–	Nonlinear EKF.
[13]	Anti-drone jamming	–	–	–	✓	–	–	–	PID-based planning.
[14]	GPS spoofing	–	✓	–	–	–	✓	–	RLDM-UAV (DQN).
[15]	Swarm anti-jamming	–	✓	–	–	–	–	✓	Decentralized MARL.
[16]	Comm. performance	–	✓	–	–	–	–	✓	Multi-param RL.
[17]	Jammer avoidance	–	–	–	✓	–	–	–	Radio-map planning.
[18]	Power allocation	–	–	✓	–	✓	–	–	Congestion game.
[19]	Large-scale networks	–	–	✓	–	✓	–	✓	Mean-field game.
[20]	UAV jammer	–	✓	–	–	–	–	–	DRQN + MDP.
Ours	GPS-denied navigation	✓	✓	–	✓	–	–	–	EKF–DRL planner.

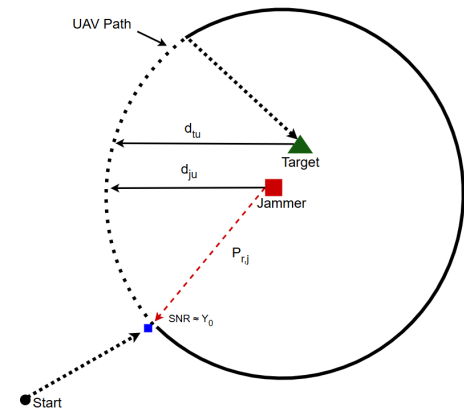
**FIGURE 1.** System model.

reinforcement learning, enabling robust target approach even under jamming conditions. Table 1 for a comparison of GPS jamming and spoofing countermeasures for UAV navigation. Table 1 summarizes the presented related works and provides a comparison of GPS jamming countermeasures for UAV navigation.

III. SYSTEM MODEL AND ASSUMPTIONS

In this work, we address the challenge of detecting and mitigating jamming attacks targeting UAVs during mission execution. Specifically, our approach focuses on enabling a UAV to reach a ground target in the presence of a hostile jammer. The key idea is to estimate the jammer transmission power and the associated path loss parameters, which are then leveraged to infer the power received from the intended target. These estimates are performed while the UAV hovers around the region affected by the jammer, prior to approaching the final destination, as illustrated in Fig. 1.

We consider a three-dimensional (3D) operational environment in which the UAV operates at position (x_u, y_u, z_u) , the GPS source is located at (z_g, y_g, z_g) . Both the jammer and the target node are located on the ground at positions (z_j, y_j) and (z_t, y_t) , respectively. The UAV obtains its positional awareness from the GPS signal S_g and navigates toward

**FIGURE 2.** UAV flight path showing distance d_{tu} , jamming power $P_{r,j}$, and SNR near y_0 .

the target. However, when a jammer is positioned near the target, it transmits a strong jamming signal S_j , which elevates interference and degrades the GPS signal-to-noise ratio (SNR) at the UAV. This degradation becomes especially severe as the UAV approaches the jammer, potentially leading to a complete loss of the GPS signal.

We assume that within the jamming region the UAV still receives the GPS signal, albeit with an SNR that approaches the system-defined threshold. Both GPS and jamming signals are modeled using the standard log-distance path loss propagation model with log-normal shadowing, expressed as:

$$P_r(d) = P_t - 10n \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma \quad (1)$$

where $P_r(d)$ is the power received at distance d (in dBm), P_t is the transmission power (in dBm), n is the path loss exponent, d is the distance between the transmitter and receiver in meters, and d_0 is a reference distance (typically set at 1 meter). X_σ is a zero-mean Gaussian random variable with standard deviation σ representing the random shadow effects.

A. UAV BEHAVIOR AT THE JAMMING BOUNDARY

As the UAV advances along its planned trajectory toward the target, it enters regions where the received signal quality progressively degrades due to the presence of an unknown jamming source. In the proposed approach, the UAV continuously monitors the SNR during its flight. When the measured SNR approaches the predefined system threshold γ_0 (i.e., $\text{SNR} \approx \gamma_0$), the UAV maintains sufficient signal quality to preserve GPS-based positioning and avoids penetrating deeper into the jammed region. Instead, it initiates a circular hovering maneuver around the target node, as illustrated in Fig. 2, in order to keep the received SNR above the threshold and ensure reliable communication.

The overall operation is divided into two sequential stages. In the first stage, the UAV flies toward the target while continuously monitoring the SNR. During this stage, the target location is assumed to be known, and GPS information remains available. Once the measured SNR drops to approximately equal the predefined threshold γ_0 , indicating proximity to the jamming boundary, the UAV transitions to a circular hovering mode.

While hovering, the UAV continues to rely on GPS to collect essential measurements and estimate key signal parameters using the EKF, as shown in Fig. 3. These parameters include the transmit power, the path-loss exponent, and the predicted received power. The estimation process is based on the total noise inferred from the SNR, which is commonly provided by communication hardware.

The second stage, also depicted in Fig. 3, corresponds to the DRL component. In this stage, the UAV enters the jammer region, where GPS signals become unreliable due to strong interference. Consequently, GPS information is no longer required. Instead, the DRL agent utilizes the EKF estimated parameters obtained in the first stage to guide the UAV by continuously updating its position toward the target.

The circular path around the target is computed using the following procedure. Let the UAV's position be (x_u, y_u, z_u) , and define the distance to the target as:

$$r = d_{tu} = \sqrt{(x_u - x_t)^2 + (y_u - y_t)^2 + (z_u - z_t)^2} \quad (2)$$

The angular velocity of the circular motion is defined as:

$$\omega = \frac{\pi}{T/c} \quad (3)$$

where T is the number of flight steps and $c \in \{1, 2\}$ indicates whether the UAV performs a half-circle or full-circle maneuver. The initial angle relative to the target is:

$$\theta_0 = \arctan 2(y_u - y_t, x_u - x_t) \quad (4)$$

Leaving $t = 0, 1, \dots, T-1$ represent the discrete time steps, the angular position at each step is computed as:

$$\theta_t = \omega t + \theta_0 \quad (5)$$

The UAV's trajectory is then defined as:

$$\begin{aligned} x_u[t] &= x_t + r \cdot \cos(\theta_t) \\ y_u[t] &= y_t + r \cdot \sin(\theta_t) \\ z_u[t] &= z_u \end{aligned} \quad (6)$$

This maneuver enables the UAV to gather a set of signal measurements over a range of angles and distances, which are used to estimate the jamming model parameters. Subsequently, these parameters are fed into the EKF and DQN modules to guide the UAV through the jammed region and reach the final destination, as detailed in the following sections.

IV. PROPOSED EKF-BASED ESTIMATION OF TRANSMIT POWER AND PATH LOSS EXPONENT

The EKF is a nonlinear extension of the Linear Kalman Filter (LKF) that accounts for system nonlinearities through first-order linearization of the underlying observation model. It recursively estimates the state mean and covariance, making it well suited for practical scenarios in which exact linearity cannot be assumed. In this work, the EKF is adopted due to its favorable trade-off between estimation accuracy and computational efficiency. The dominant nonlinearity in the received power model arises from the logarithmic path-loss relationship, which is smooth and can be accurately approximated via first-order linearization within the UAV's operational region. Although alternative estimators such as Unscented Kalman Filters (UKF) or Particle Filters (PF) are capable of handling stronger nonlinearities or multimodal uncertainty, their substantially higher computational and memory requirements limit their applicability in real-time, resource-constrained UAV platforms. Under the assumptions of stationary jamming sources and Gaussian measurement noise, the EKF achieves reliable convergence with significantly lower complexity, making it an appropriate and practical choice for the proposed EKF-DRL navigation framework [24], [25], [26].

In the context of autonomous UAV navigation in GPS-denied environments, it is essential to estimate both the jammer's transmission power P_t and the path loss exponent n , as these parameters determine the strength of the received signal and are fundamental to inferring spatial relationships. In this work, we adopt an EKF-based estimation framework, which operates while the UAV executes a circular hovering trajectory around the jamming region. This maneuver allows for iterative refinement of the EKF's estimates of P_t and n , as detailed in Algorithm 1.

Let the UAV follow a trajectory defined by coordinates (x_u, y_u, z_u) , while the jammer and known target are located at fixed positions (x_j, y_j) and (x_t, y_t) , respectively. At each hovering point, the UAV computes the Euclidean distance of the target as in equation (2).

The received signal power from the jammer at the UAV's position is modeled using the logarithmic path loss

model [26], [27]:

$$P_{r,j}^{\text{meas}} = P_t - 10 n \log_{10}(d_{ju}) + \mathcal{N}(0, \sigma^2) \quad (7)$$

where the distance from jammer to UAV is defined as:

$$d_{ju} = \sqrt{(x_u - x_j)^2 + (y_u - y_j)^2 + (z_u - z_j)^2} \quad (8)$$

where $\mathcal{N}(0, \sigma^2)$ denotes zero-mean Gaussian noise with variance σ^2 . This noisy measurement is provided as input to the EKF update mechanism.

In our model, we use d_{tu} , the distance of the UAV to the target, as part of the observation function to allow the EKF to virtually project the jammer to the target location. This virtual alignment enables the filter to estimate the transmit power at the target's position, under the assumption that the jammer is co-located with the target. Recomputing the power received at this virtual target location, the UAV can estimate the received power from the jammer had it been placed at the target. This forms the basis for future guidance decisions. The EKF observation function is defined as follows:

$$h(\hat{p}_t, d_{tu}) = \hat{p}_t - 10n \log_{10}(d_{tu}) \quad (9)$$

where $\hat{p}_t$ represents the estimated transmission power. The corresponding Jacobian simplifies to the following:

$$H = \frac{\partial h}{\partial \hat{p}_t} = 1 \quad (10)$$

The EKF operates through standard prediction and correction steps:

$$\text{Predict: } \hat{p}_{t|k|k-1} = \hat{p}_{t|k-1|k-1} \quad (11)$$

$$P_{k|k-1} = P_{k-1|k-1} + Q \quad (12)$$

$$\text{Update: } K_k = \frac{P_{k|k-1}H}{HP_{k|k-1}H + R} \quad (13)$$

$$\hat{p}_{t|k|k} = \hat{p}_{t|k|k-1} + K_k (z_k - h(\hat{p}_{t|k|k-1}, d_{tu})) \quad (14)$$

$$P_{k|k} = (1 - K_k H)P_{k|k-1} \quad (15)$$

Once the estimate of the transmission power $\hat{P}_t$ is updated, the path loss exponent $\hat{n}$ is inferred using a signal-to-noise ratio SNR_t at the target:

$$\hat{n} = \frac{\hat{P}_t - (\text{SNR}_t + N_0)}{10 \log_{10}(d_{tu})} \quad (16)$$

where N_0 represents the thermal noise floor in dBm. Using the estimated $\hat{n}$, the UAV can update its estimate distance to the jammer:

$$\hat{d}_{ju} = d_0 \cdot 10^{\frac{\hat{P}_t - P_{r,j}^{\text{meas}}}{10\hat{n}}} \quad (17)$$

With $\hat{P}_t$ and $\hat{n}$ in hand, the UAV can estimate the power received from the jammer at the target position:

$$\hat{P}_{r,t} = \hat{P}_t - 10\hat{n} \log_{10}(d_{tu}) \quad (18)$$

Algorithm 1 EKF-Based Power Estimation

Require: UAV trajectory (x_u, y_u, z_u) , target position (x_t, y_t, z_t)

Ensure: Estimated parameters $(\hat{P}_t, \hat{n}, \hat{P}_{r,t})$

Initialization:

$\hat{P}_t \leftarrow P_{t,\text{init}}$

$P \leftarrow \sigma_{P_t}^2$

$Q, R \leftarrow$ process and measurement noise covariances

$N_0 \leftarrow$ noise floor, $\epsilon \leftarrow$ convergence threshold

$\text{SNR}_t \leftarrow$ system threshold

for each UAV position (x_u, y_u, z_u) **do**

$d_{tu} \leftarrow \sqrt{(x_u - x_t)^2 + (y_u - y_t)^2 + (z_u - z_t)^2}$

Obtain measurement:

$P_{r,j}^{\text{meas}} \leftarrow P_t - 10n \log_{10}(d_{ju}) + \mathcal{N}(0, \sigma^2)$

EKF Prediction Step:

$\hat{P}_t^- \leftarrow \hat{P}_t$

$P^- \leftarrow P + Q$

EKF Update Step:

$H \leftarrow 1$

$K \leftarrow \frac{P^- H}{H P^- H + R}$

$\hat{P}_t^- \leftarrow \hat{P}_t^- + K (P_{r,j}^{\text{meas}} - (\hat{P}_t^- - 10n \log_{10} d_{tu}))$

$P \leftarrow (I - KH)P^-$

Parameter Estimation:

$\hat{n} \leftarrow \frac{P_t - (\text{SNR}_t + N_0)}{10 \log_{10} d_{tu}}$

$\hat{d}_{ju} \leftarrow 10^{(\hat{P}_t - P_{r,j}^{\text{meas}})/(10\hat{n})}$

$\hat{P}_{r,t} \leftarrow \hat{P}_t - 10\hat{n} \log_{10} d_{tu}$

$\hat{P}_{r,j} \leftarrow \hat{P}_t - 10\hat{n} \log_{10} \hat{d}_{ju}$

$\text{SNR}_t \leftarrow \hat{P}_{r,t} - N_0$

if $|P_{r,j}^{\text{meas}} - \hat{P}_{r,j}| \leq \epsilon$ **then**

break

end if

end for

return $(\hat{P}_t, \hat{n}, \hat{P}_{r,t})$

Based on this estimated distance, the UAV also recalculates the expected received power from the jammer at its current location:

$$\hat{P}_{r,j} = \hat{P}_t - 10\hat{n} \log_{10}(\hat{d}_{ju}) \quad (19)$$

This estimate enables computation of the expected signal-to-noise ratio at the UAV from the target:

$$\text{SNR}_t = \hat{P}_{r,t} - N_0 \quad (20)$$

The EKF loop continues until convergence, defined as when the received power estimation error satisfies the following:

$$\varepsilon_P = |P_{r,j}^{\text{meas}} - \hat{P}_{r,j}| \leq \epsilon \quad (21)$$

where ϵ is a predefined threshold that balances convergence accuracy and computational cost.

Finally, the estimated received power at the target $\hat{P}_{r,t}$ and the refined path-loss exponent $\hat{n}$ are used as state inputs to a DQN-based reinforcement learning model, as illustrated in the system block diagram in Fig. 3. This

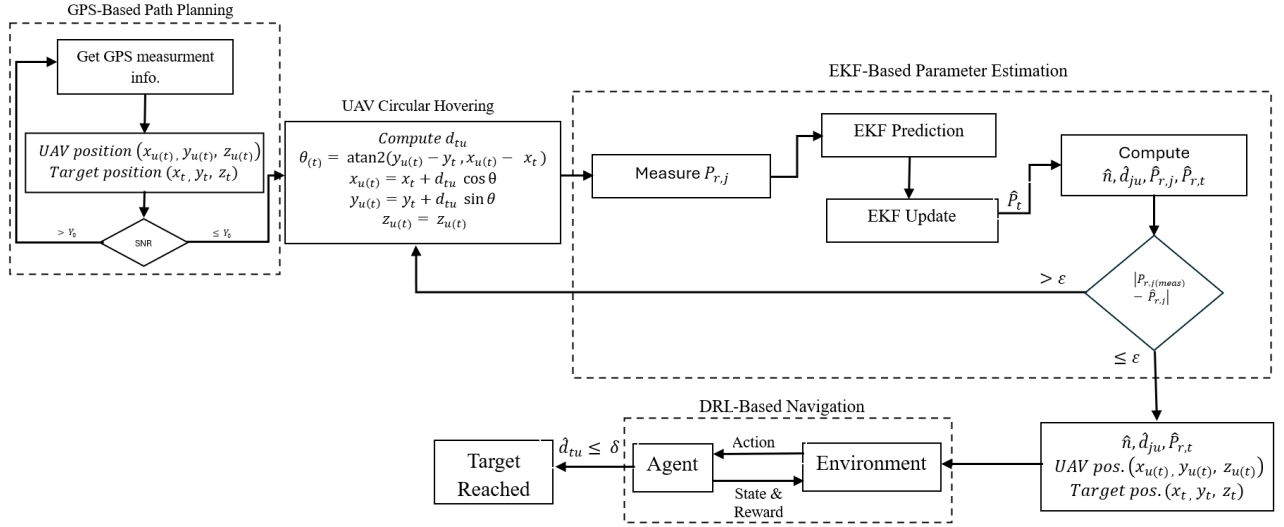


FIGURE 3. Proposed EKF-DRL-based UAV navigation in GPS-denied environments using EKF estimates of jammer power $\hat{P}_t$, path loss exponent $\hat{n}$, and received power at the target $\hat{P}_{r,t}$ to construct the agent's observation state.

integration enables the UAV to autonomously adjust its trajectory toward the target under GPS-denied conditions by leveraging the predicted signal strength as a reward-guided navigation metric. The DQN policy exploits these estimates to determine optimal flight actions that maximize signal gain and estimation reliability, thereby achieving resilient path planning in adversarial RF environments.

V. REINFORCEMENT LEARNING

Reinforcement Learning is a powerful framework that enables autonomous agents to learn optimal behavior through interaction with their environment. In RL, an agent observes the state of the environment, selects actions, and receives feedback in the form of rewards, using this experience to improve its policy over time. DQN, a class of DRL methods, approximates the action-value function $Q(s, a)$ using neural networks, allowing scalable learning in high-dimensional or partially observable environments [28], [29], [30].

A. PROPOSED DQN-BASED NAVIGATION FOR UAVS IN GPS-JAMMING ENVIRONMENTS

The proposed navigation framework employs a DQN-based reinforcement learning agent to guide a UAV in environments without GPS access. The UAV relies exclusively on RF-based estimations, using EKF outputs as state observations to approach a fixed target. The EKF module provides three critical estimates: the jammer's transmit power $\hat{P}_t$, the path loss exponent $\hat{n}$, and the estimated received power at the target node $\hat{P}_{r,t}$. These are continuously updated according to the location of the UAV and are used to construct the observation state of the agent.

Navigation is modeled as a Markov Decision Process (MDP), defined by (S, A, T, R, γ) :

- S is the set of observed states, consisting of normalized UAV coordinates (x_u, y_u, z_u) , normalized distance to the target, and normalized EKF-estimated received power.
- A is the discrete action space with 8 movement directions in the xy plane: $\{N, NE, E, SE, S, SW, W, NW\}$.
- T represents the transition dynamics, governed by UAV kinematics and the signal-aware altitude scaling.
- R is the reward function that encourages proximity to the target, higher signal quality, and successful completion of the task.
- γ is the discount factor to balance immediate and long-term returns.

At each time step t , the UAV computes the 3D distance $d_{tu}(t)$ to the target and updates the received power estimate using the EKF-based model in equation (18).

The agent selects an action a_t using an ϵ -greedy policy applied to $Q_\theta(s_t)$. After executing the action, the UAV's x and y positions are updated based on the chosen direction, while the altitude is scaled according to the ratio of current-to-initial distance from the target. The reward function used to guide learning is defined as:

$$r_t = w_1 \left(\frac{1}{d_t + \epsilon} \right) + w_2 \cdot \hat{P}_{r,t}(t) + w_3 \cdot \mathbb{I}(d_t < \delta), \quad (22)$$

where the first term encourages proximity to the target via inverse distance weighted by w_1 ; the second term, scaled by w_2 , aligns the agent's movement with regions of stronger estimated signal power based on EKF output; and the third term assigns a fixed success bonus w_3 when the UAV reaches within a predefined distance threshold δ of the target. The indicator function $\mathbb{I}(\cdot)$ activates this bonus when the condition is met. The Q-network is updated through mini-batch training using the target value:

$$Y_t = r_t + \gamma \max_{a'} Q_{\theta^-}(s_{t+1}, a'), \quad (23)$$

Algorithm 2 DRL-Based UAV Navigation

Require: EKF estimates, UAV position (x_u, y_u, z_u) , Target (x_t, y_t, z_t)

Ensure: Optimized policy π^*

Initialize replay buffer D , DQN Q_θ , target network Q_{θ^-} , ϵ
for episode = 1 to M **do**

Reset UAV, get initial state s_0 , $R_{\text{episode}} \leftarrow 0$

for $t = 1$ to T_{max} **do**

$d_{\text{tu}}(t) \leftarrow \|(x_u, y_u, z_u) - (x_t, y_t, z_t)\|_2$

Update $\hat{P}_{r,t} \leftarrow \hat{P}_t - 10\hat{n} \log_{10}(d_{\text{tu}}(t))$

$s_t \leftarrow [x_u, y_u, z_u, \hat{P}_{r,t}, d_{\text{tu}}(t), \Delta \hat{P}_{r,t}]$

Select a_t via ϵ -greedy from $Q_\theta(s_t)$

Update position (x_u, y_u) , altitude z_u

Compute reward r_t using distance and power

Store (s_t, a_t, r_t, s_{t+1}) in D

if enough samples **then**

Train Q_θ on mini-batch from D

Update target network Q_{θ^-}

end if

Decay ϵ

if $d_{t+1} < \delta$ **then**

break

end if

end for

end for

return Q_θ

and the update rule is derived from the Bellman equation:

$$\mathcal{L}(\theta) = (Y_t - Q_\theta(s_t, a_t))^2. \quad (24)$$

Training proceeds for multiple episodes until the UAV reaches a distance threshold from the target or exhausts the allowed number of steps. This learning-based scheme, summarized in Algorithm 2, enables GPS-free autonomous navigation through real-time signal feedback and policy optimization.

VI. SIMULATION AND RESULTS

A. SIMULATION SETUP

To evaluate the proposed EKF-assisted DRL navigation framework, we perform a series of numerical simulations using Python. Unless otherwise specified, noise terms are assumed to be Gaussian, and all power are in dBm. The jammer transmit power P_t is initialized to -20 dBm with the EKF process and the measurement noise covariances set to $Q = 10^{-7}$ and $R = 110$, respectively. The state error covariance is initialized to $P = 0.2$, with a nominal path loss exponent set to $n = 2$. The initial coordinates of the UAV are $(50, 95, 50)$ and the transmit power of the jammer is $P_t = -30$ dBm. The noise floor is fixed to $N_0 = -90$ dBm. The target node is located at $(100, 70, 0)$, while the jammer is placed at $(95, 50, 0)$. The UAV performs an outward path planning stage followed by a circular hovering maneuver. The circular radius is determined from the Euclidean distance

TABLE 2. Effect of jamming power error threshold ϵ on EKF convergence performance.

ϵ	Stopping Step	n Error	P_t Error
0.4302	28	2.15×10^{-2}	4.29×10^{-1}
0.0684	36	3.29×10^{-3}	6.57×10^{-2}
0.0066	41	3.33×10^{-4}	6.65×10^{-3}
0.0028	116	1.38×10^{-4}	2.76×10^{-3}
0.00050	327	2.51×10^{-5}	5.02×10^{-4}

between the UAV and the target with the $T = 5000$ hover steps. during hovering of the UAV. EKF predictions are terminated once the residual error satisfies $|P_{r,j}^{\text{meas}} - \hat{P}_{r,j}| \leq \epsilon$.

B. EKF ESTIMATION PERFORMANCE ANALYSIS

The results in Fig. 4 illustrate the performance of the EKF parameter estimation process under varying jamming power error thresholds ϵ , with the transmit power fixed at $P_t = -30$ dBm and $\epsilon \in \{0.5, 0.08, 0.02, 0.004, 0.001\}$. The four subplots correspond to: (a) the EKF-estimated jammer transmit power, (b) the received jamming power error at the UAV, (c) the transmit power estimation error, and (d) the path-loss exponent estimation error.

Each curve represents the convergence trajectory for a different ϵ , with the terminal point marked by an 'X' symbol. As summarized in Table 2, a higher threshold (for example, $\epsilon = 0.4302$) results in the fastest convergence—requiring only 28 steps with a final n error of 2.15×10^{-2} and P_t error of 4.29×10^{-1} . In contrast, a very small threshold (e.g., $\epsilon = 0.00050$) demands 327 steps but achieves a much smaller final n error of 2.51×10^{-5} and P_t error of 5.02×10^{-4} .

The observed trends remain consistent across all tested values: decreasing ϵ slows the UAV's adaptation to the target's signal environment, thereby prolonging the hovering phase before parameter estimates stabilize. On the other hand, moderate ϵ values (e.g., $\epsilon = 0.0066$) facilitate faster convergence within 41 steps while still maintaining acceptable accuracy, making them suitable for balancing real-time navigation with estimation precision. These results confirm that ϵ serves as a tunable parameter that governs the trade-off between accuracy and convergence speed in the EKF-DRL navigation framework.

C. REINFORCEMENT LEARNING CONFIGURATION

The proposed DQN architecture employs a multi-layer perceptron with four fully connected layers to process a 6-dimensional state vector consisting of EKF estimates and UAV positions. The network comprises an input layer (6 neurons), two hidden layers (128 neurons each with ReLU activation), a third hidden layer (64 neurons with ReLU), and an output layer (8 neurons) representing discrete actions. This configuration balances learning capacity with real-time computational efficiency.

Training utilizes experience replay with a memory size of 10,000 and mini-batch gradient descent (batch size 64). An ϵ -greedy strategy decays from 1.0 to 0.05, with a discount

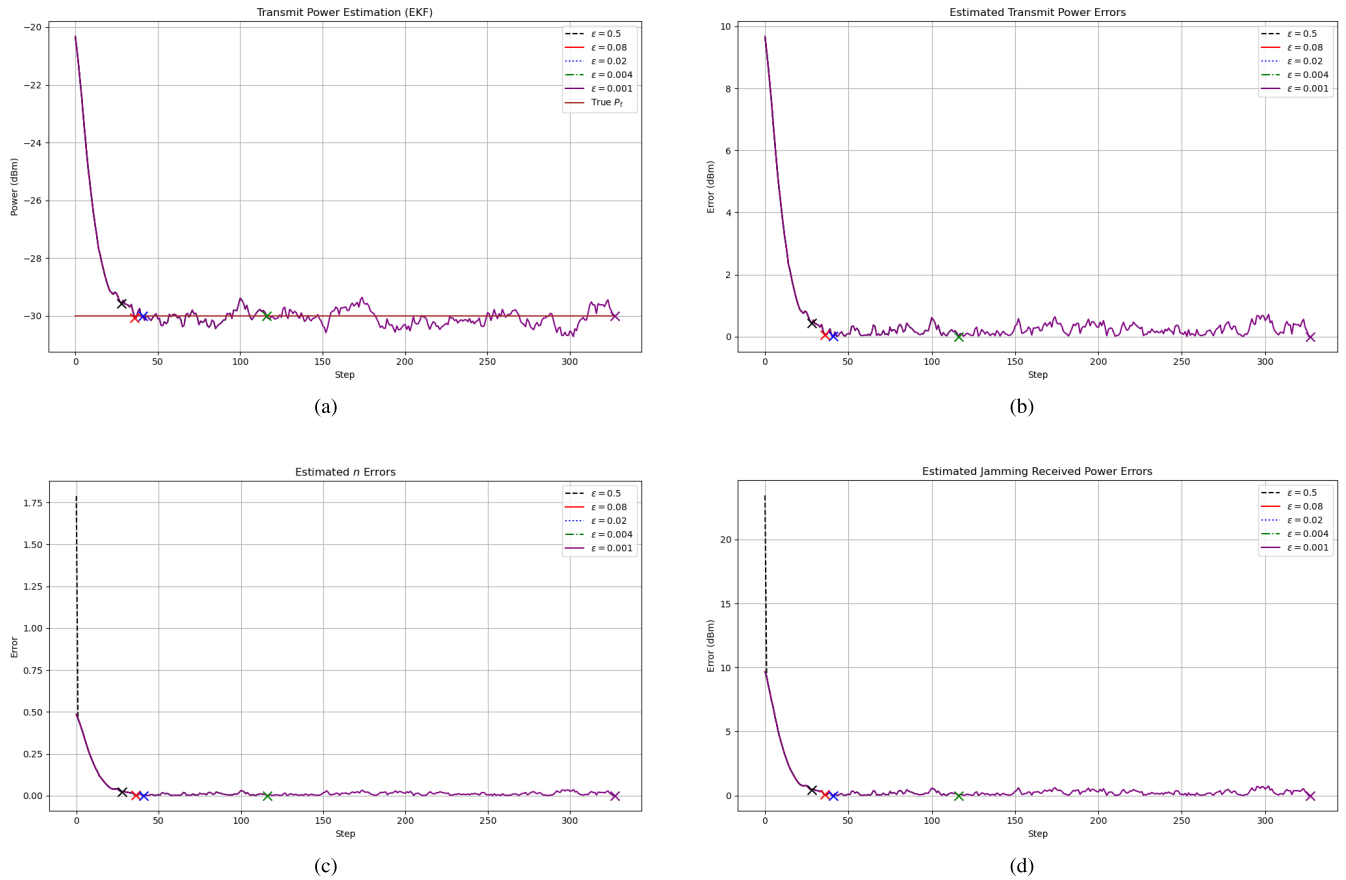


FIGURE 4. EKF estimation performance across different ϵ thresholds. (a) Transmit power estimation (dBm), (b) transmit power estimation error (dBm), (c) path-loss exponent estimation error, and (d) received jamming power error at UAV for $\epsilon \in \{0.5, 0.08, 0.02, 0.004, 0.001\}$ with $P_t = -30$ dBm.

TABLE 3. DRL model implementation.

Component	Parameter	Value
Network	Input Layer	6 neurons
	Hidden Layer 1	128 neurons
	Hidden Layer 2	128 neurons
	Hidden Layer 3	64 neurons
	Output Layer	8 neurons
	Activation	ReLU
Training	Memory Size	10,000
	Batch Size	64
	Episodes	200
	Max Steps	300
Hyperparameters	Learning Rate	10^{-1} – 10^{-4}
	Discount (γ)	0.5
	Optimizer	Adam
	Initial ϵ	1.0
	Final ϵ	0.05
	ϵ Decay	0.995
Reward Weights	w_1 (Distance)	1.0
	w_2 (Power)	0.5
	w_3 (Success)	10.0

factor of $\gamma = 0.5$. The network is optimized using Adam with the MSE loss between predicted and target Q-values. Training is conducted over 200 episodes, with up to 300 steps per episode, and learning rates are evaluated from 10^{-1} to

10^{-4} . Table 3 summarizes the network architecture, state representation, training configuration, hyperparameters, and reward weights.

D. EKF-DRL PERFORMANCE ANALYSIS

In each trial with different learning rates, we record the following performance metrics: (i) cumulative episode reward, (ii) the number of steps required for the UAV to reach the target, and (iii) the success rate and final position error at the end of each episode. The results are further visualized through convergence plots of $\hat{P}_t$ versus P_t , $\hat{P}_{r,t}$ versus $P_{r,t}$, path-loss exponent estimation, and 2-D/3-D UAV trajectory evolution. The DRL performance is evaluated by jointly considering convergence stability, navigation accuracy, and efficiency. The cumulative reward of each episode is defined as:

$$R_{\text{episode}} = \sum_{t=1}^T r_t \quad (25)$$

while the number of steps is expressed as:

$$S_{\text{episode}} = \begin{cases} t, & \text{if the target is reached at step } t \\ T_{\text{max}}, & \text{otherwise} \end{cases} \quad (26)$$

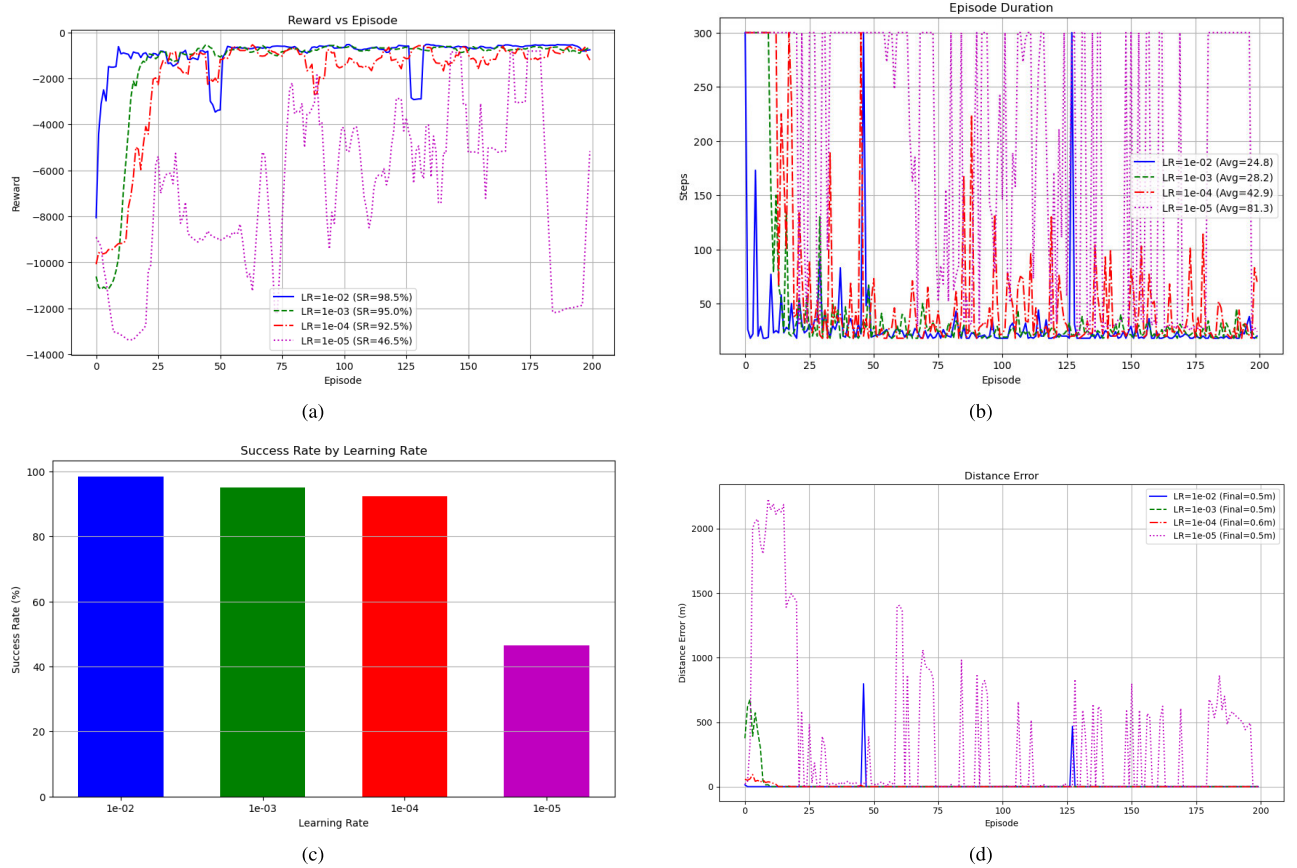


FIGURE 5. Performance comparison of the proposed DRL navigation model under different learning rates: (a) episode reward progression, (b) steps required to reach target per episode, (c) success rate comparison across learning rates, and (d) final distance error analysis.

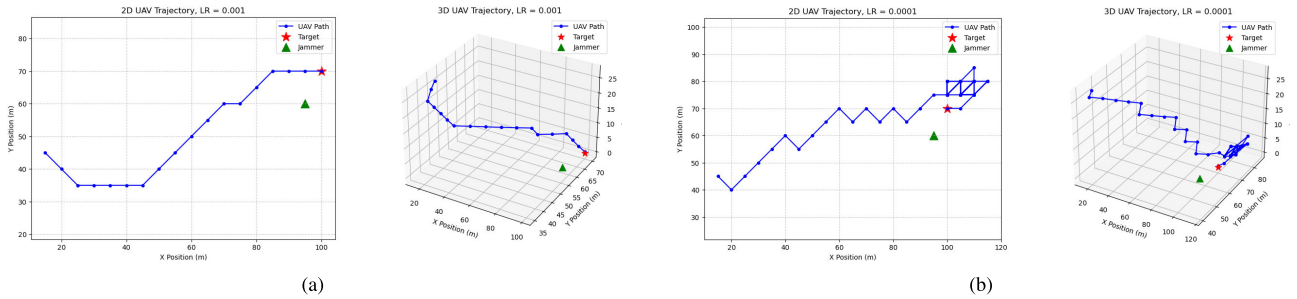


FIGURE 6. UAV trajectories under different learning rates (LR): (a) $LR = 10^{-3}$ and (b) $LR = 10^{-4}$. The plots highlight the effect of the learning rate on convergence speed and final positioning accuracy.

TABLE 4. Performance of DRL model under different learning rates.

Learning Rate	SR (%)	Avg. Steps	Dist. Error (m)
1×10^{-2}	98.5	24.8	0.5
1×10^{-3}	95.0	28.2	0.5
1×10^{-4}	92.5	42.9	0.6
1×10^{-5}	46.5	81.3	0.5

where $T_{\max}$ denotes the maximum number of steps allowed per episode. The final position error is measured as the Euclidean distance between the UAV's terminal position (x_u, y_u, z_u) and the target (x_t, y_t, z_t) :

$$e_{\text{pos}} = \|(x_u, y_u, z_u) - (x_t, y_t, z_t)\|_2 \quad (27)$$

The success rate (SR) is finally computed as:

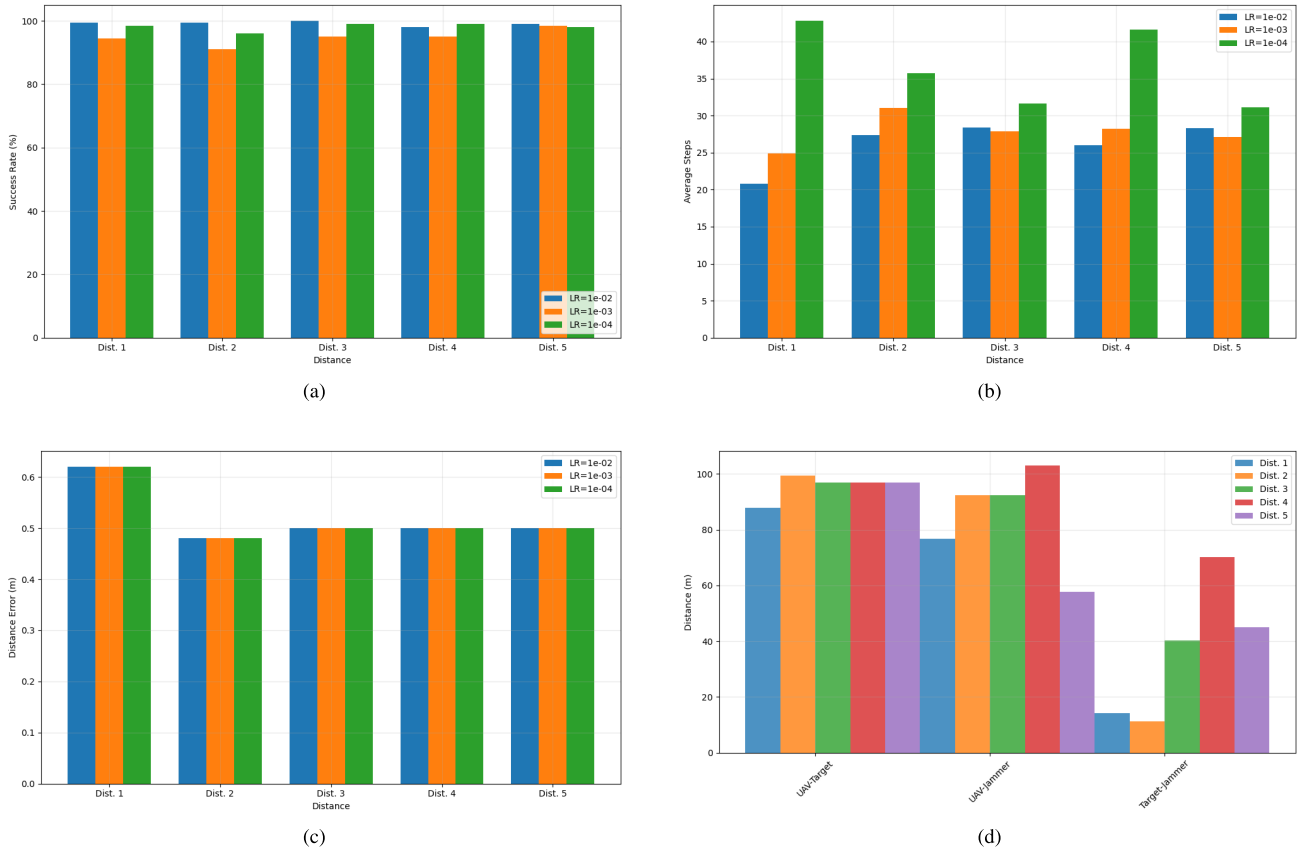
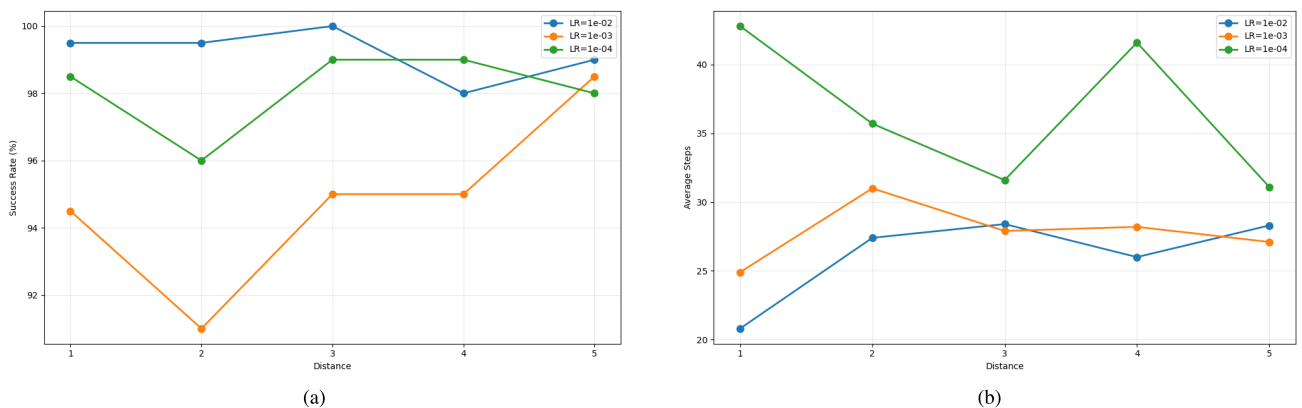
$$SR = \frac{N_{\text{success}}}{N_{\text{episodes}}} \times 100\% \quad (28)$$

where N_{success} and N_{episodes} denote the number of successful and total episodes, respectively.

Fig. 5 compares the navigation performance of the DRL framework at different learning rates. As observed, $LR = 10^{-2}$ achieves the best overall trade-off, with a success rate of 98.5%, an average of 24.8 steps to reach the target, and a final distance error of 0.5 m, thus demonstrating both rapid convergence and high precision. A slightly lower success rate of 95.0% is obtained with $LR = 10^{-3}$, which nevertheless preserves a similar convergence speed

TABLE 5. EKF-DRL performance under different jamming transmit powers.

Learning Rate	$P_t = -10$ dBm			$P_t = -15$ dBm			$P_t = -25$ dBm		
	SR (%)	Avg Steps	dist. Error (m)	SR (%)	Avg Steps	dist. Error (m)	SR (%)	Avg Steps	dist. Error (m)
1×10^{-1}	4.5	44.0	0.55	3.0	58.5	0.54	3.0	82.5	0.50
1×10^{-2}	99.0	25.4	0.48	98.0	23.6	0.48	97.5	29.7	0.48
1×10^{-3}	97.5	23.4	0.48	97.0	22.7	0.48	88.0	35.9	0.48
1×10^{-4}	99.0	34.1	0.48	94.5	33.2	0.48	99.8	30.8	0.48

**FIGURE 7.** Model performance comparison across different distances ($P_t = -25$ dBm, 200 episodes).**FIGURE 8.** Success rate and average steps of EKF-DRL across different distances for varying learning rates.

and accuracy. In contrast, $LR = 10^{-5}$ produces the weakest performance, with only a success rate of 46.5%,

a significantly longer convergence horizon of 81.3 steps, and reduced reliability despite a small final error of 0.5 m.

Finally, $LR = 10^{-4}$ provides a more balanced but slower performance, reaching 92.5% success with an average of 42.9 steps and a final error of 0.6 m. Figures 6(a)–(b) further illustrate UAV trajectories under representative learning rates.

Table 4 presents the results, showing that $LR = 10^{-2}$ at $P_t = -30$ dBm provides the optimal balance between stability, precision, and convergence efficiency within the EKF–DRL navigation framework. Table 5 further demonstrates the robustness of the proposed EKF–DRL framework under varying jamming transmit powers. Learning rates of 1×10^{-2} and 1×10^{-4} consistently achieve high success rates, ranging from 97.5% to 99.8%, with distance errors of 0.48–0.50 m, regardless of interference intensity. The stability of the distance error between the UAV and the target is primarily attributed to the EKF. In the first stage, the EKF explicitly estimates jammer-related parameters, including transmit power and path-loss components, and passes these refined estimates to the DRL agent. Consequently, variations in the jammer transmit power are largely compensated for during the EKF estimation process. Because the DRL model relies on EKF-refined estimates rather than raw received power measurements, its performance is relatively insensitive to changes in jammer transmit power. This explains why the final distance error remains nearly constant across different TX power levels. The slightly improved performance observed at higher jammer transmit powers can be attributed to enhanced signal observability, which improves EKF convergence and leads to more accurate parameter estimation.

Figure 7 and Figure 8 present the performance evaluation of the proposed EKF–DRL framework under multiple spatial configurations at a transmit power of $P_t = -25$ dBm, using 200 training episodes. Figure 7(a) illustrates the achieved success rates across five distinct distance scenarios, while the corresponding geometric relationships among the UAV, the target, and the jammer are depicted in Figure 7(d). Across all evaluated configurations, the framework consistently achieves success rates exceeding 97%, demonstrating strong robustness to variations in target–jammer geometry.

As shown in Figure 7(d), the considered scenarios cover a wide range of deployment conditions, including both short and extended UAV–jammer separations, as well as varying target–jammer distances. Specifically, the UAV–target distance spans from 87.75 m to 99.50 m, the UAV–jammer distance ranges between 57.66 m and 103.08 m, and the target–jammer distance varies from 11.18 m to 70.18 m. Despite these pronounced differences in spatial geometry, the proposed framework maintains stable performance across all scenarios.

Figure 7(c) shows that the localization distance error remains tightly bounded within the range of 0.48–0.63 m for all evaluated positions, confirming the accuracy and reliability of the EKF-based estimation process even under strong jamming conditions. The convergence behavior illustrated in Fig. 7(b) indicates minor sensitivity to the learning rate, where learning rates of 10^{-2} and 10^{-3} achieve

TABLE 6. Performance over 400 episodes at $P_t = -10$ dBm.

Learning Rate	SR (%)	Avg Steps	Dist. Error (m)
1×10^{-1}	4.2	74.0	244.51
1×10^{-2}	99.2	21.9	2.60
1×10^{-3}	99.0	21.8	5.64
1×10^{-4}	98.8	25.8	2.48

faster convergence compared to 10^{-4} , while maintaining comparable steady-state performance.

Overall, the results presented in Figure 7 and Figure 8 highlight the stability and reliability of the EKF–DRL framework for UAV navigation across diverse deployment settings, demonstrating its ability to sustain high success rates, accurate localization, and consistent learning behavior under varying spatial configurations.

Extending training from 200 to 400 episodes, as summarized in Table 6, further improves performance, particularly under the strongest jamming condition ($P_t = -10$ dBm). In this scenario, $LR = 10^{-2}$ attains a 99.2% success rate with an average of 21.9 steps, and $LR = 10^{-4}$ also demonstrates reliable performance. These results confirm that the EKF–DRL framework ensures both robust navigation and enhanced convergence efficiency, even under severe jamming conditions.

E. COMPARATIVE PERFORMANCE DISCUSSION

A comparison of recent UAV resilience techniques is presented in Table 7. In general, most existing methods focus on detection or recovery under specific attack types, often requiring external sensors, prior environmental data, or cooperative multi-agent coordination. These approaches tend to increase system complexity or reduce adaptability in dynamic conditions. In contrast, the proposed EKF–DQN framework integrates environment-aware estimation through EKF with adaptive decision-making using DQN, enabling autonomous operation with minimal external dependency. The hybrid design achieves high localization accuracy (≈ 0.5 m error) and a 98.5% success rate across different jamming levels, demonstrating an effective balance between estimation precision, convergence speed, and navigation stability in GPS-denied environments.

F. LIMITATIONS

In this study, both the jammer and the target are modeled as stationary nodes. This assumption yields relatively stable channel conditions during the estimation phase, enabling reliable convergence of the EKF for estimating the jammer transmit power and the path-loss exponent.

The proposed framework further assumes that the jammer is stationary in space and exhibits quasi-stationary transmission characteristics during the EKF estimation interval. Specifically, the jammer's transmit power and waveform statistics are assumed to remain constant or to vary slowly over the circular hovering maneuver used for parameter estimation. This assumption allows the EKF to exploit

TABLE 7. Comparative analysis.

Method	Approach	Threat Scenario	Performance Metrics	Results & Accuracy	Limitations
[10]	Deep Q-Network for spoofing detection	GPS spoofing (random/replay/stealth)	Detection time < 23 ms; Completion rate	98.8% detection accuracy, 91.5% mission completion	Requires camera input; Focused on detection only
[13]	Passive jamming-based navigation	Full-spectrum jamming	Computational overhead $\approx 70\%$	100% target reachability in tests	No learning capability; No parameter estimation
[15]	Multi-Agent RL (Dec-POMDP)	Cooperative anti-jamming, dynamic channels	Packet Loss Rate reduction vs. SARL; Training episodes = 500	Enhanced multi-UAV robustness	Multi-UAV only; High computational complexity
[23]	RSS trilateration + Artificial Potential Field	Cyber/GPS attacks	Convergence iterations = 100; Success rate up to 98%	98% mission success rate	Limited adaptability; No jamming awareness
Proposed	EKF-based jammer estimation + DQN navigation	GPS-denied, jamming environment	Distance error ≈ 0.5 m; Average steps ≈ 25	98.5% success rate; Robust across jamming levels	assuming stationary target and jammer positions; Single-agent configuration

predictable path-loss relationships for stable parameter convergence. In contrast, highly adaptive jammers with rapidly time-varying transmit power, duty cycles, or waveform characteristics may violate this assumption and degrade EKF convergence performance. Addressing such non-stationary jamming scenarios would require extending the EKF state model to capture time-varying transmission dynamics or adopting adaptive filtering strategies, which is beyond the scope of this work and left for future investigation.

If both the jammer and the target are mobile, the EKF formulation will need to be extended to include additional state variables, such as velocity and acceleration components in three dimensions. In this case, the filter will not only estimate transmission parameters but also track the jammer's trajectory, leading to increased state dimensionality and higher computational complexity, as well as greater memory requirements and energy and time overhead for the EKF-DQN framework.

Similarly, if the target is mobile, the UAV can no longer rely on static circular hovering for parameter estimation. The DRL policy would need to be redesigned to operate in a non-stationary environment. In particular, the reward function would need to jointly account for trajectory tracking accuracy and received power estimation, enabling the UAV to adaptively follow a moving target while maintaining resilience to jamming interference.

From an implementation perspective, the proposed EKF-DRL framework can be deployed on embedded platforms, such as a Raspberry Pi integrated with a Software-Defined Radio (SDR) for real-time signal acquisition. The main challenges in such deployments include ensuring low-latency DRL inference and efficient EKF processing under constrained computational resources, particularly with respect to memory usage, power consumption, and execution time. Potential mitigation strategies include the use of lightweight neural network architectures, model quantization, and edge-cloud cooperative processing to balance computational load and real-time performance.

VII. CONCLUSION

In this work, we proposed an EKF-DRL framework for robust UAV navigation in GPS-jamming environments. The EKF was employed to estimate critical jamming-related parameters, including transmit power and the path-loss exponent, which were subsequently integrated into a DRL-based navigation policy. Simulation results demonstrated that the accuracy of EKF estimation is highly dependent on the jamming power error threshold ϵ . Specifically, smaller values of ϵ resulted in significantly reduced estimation errors for both n and P_t , with errors decreasing from 1.89×10^{-2} and 3.77×10^{-1} at $\epsilon = 0.3846$ to 4.84×10^{-5} and 9.66×10^{-4} at $\epsilon = 0.00094$, respectively. Although finer thresholds required more hovering steps to achieve convergence, they enabled highly accurate parameter estimation, which in turn improved the reliability of the DRL-based navigation strategy. Overall, the proposed dual-purpose framework not only mitigates the adverse effects of GPS jamming but also exploits interference as a source of actionable information, thereby enabling UAVs to achieve precise and intelligent navigation in contested environments.

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